

Pedro Delgado Henriques

Software Engineer | C# / .NET · C++ · Full-Stack Web | Microservices & iGaming

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SUMMARY

Software Engineer with 5+ years of experience across backend, full-stack web and game development. Shipped casino slot games in C++ to production at Firepot Gaming and delivered Magento e-commerce stores with ERP integrations at Bis2Bis. Currently specializing in .NET systems architecture (FIAP postgraduate): ASP.NET Core microservices, RabbitMQ, Redis, Docker and Kubernetes. Strong in performance-sensitive code, math-heavy problems (RNG, RTP, probability) and clean, tested architecture. B.Sc. in Software Engineering. Advanced English. Open to remote and hybrid roles.

CORE SKILLS

- **Languages:** C#, C++, C, PHP, Python, JavaScript, SQL, HLSL
- **Backend & .NET:** .NET 8, ASP.NET Core, Minimal APIs, Entity Framework Core, REST APIs, Microservices, Clean Architecture, DDD, SOLID, Design Patterns, xUnit, Integration Testing
- **Messaging, Cloud & DevOps:** RabbitMQ, MassTransit, Redis, Kong API Gateway, Docker, Docker Compose, Kubernetes, Terraform, AWS (Lambda, LocalStack), CI/CD, GitHub Actions, Prometheus, Grafana
- **Databases:** SQL Server, MySQL, MongoDB, Redis, SQLite; data modeling with ORM and UML
- **Game Development:** C++ on proprietary engines, slot mathematics (RNG, RTP, volatility, paytables, Monte-Carlo simulation), Game Design Documents, performance optimization, Unity, URP/HLSL shaders, Editor tooling
- **Web & E-commerce:** PHP, Magento, Laravel, React, Node.js, jQuery, HTML5, CSS3, SASS/LESS, SOAP/XML integrations, ERP integration, technical SEO
- **Security:** JWT, OAuth2, AES, Diffie-Hellman, bcrypt, XSS prevention, LGPD/GDPR awareness
- **AI-assisted development:** daily use of coding assistants (Claude Code, GitHub Copilot, ChatGPT), with critical review and validation through automated tests and CI
- **Ways of working:** Git, GitHub, GitLab, Scrum, Kanban, Jira, Trello, code review, technical documentation

PROFESSIONAL EXPERIENCE

C++ Game Developer (Mid-Level)

Dec 2024 – Jun 2026

Firepot Gaming — iGaming / casino slot machines

- Developed and shipped casino slot machine games in pure C++ on a proprietary in-house engine; titles live in production at firepotgaming.com/slots.
- Implemented gameplay features — bonus rounds, free spins and cascading reels — integrated with existing game systems.
- Designed slot math models for multiple volatility profiles (high to low variance) and tuned RTP to specification.
- Refactored critical engine modules (symbol rendering, RNG, animation) and removed performance bottlenecks: lower memory usage, higher runtime FPS and shorter loading times.
- Migrated legacy titles across engine versions, ensuring compatibility and stability in production.
- Authored complete Game Design Documents (mechanics, payable, screen flow, bonus rules); Git branching integrated with a CI/CD pipeline.

Programming Instructor — Python

Jun 2024 – Dec 2024

Ensina Mais

- Taught Python to beginner classes: programming logic, data structures and software development best practices.
- Created course materials and hands-on exercises; tracked each student's progress individually.

Full-Stack / E-commerce Developer (Junior → Mid-Level)

Nov 2021 – Jun 2024

Bis2Bis E-commerce

- Promoted from Junior to Mid-Level, progressively owning more complex projects and technical decisions.
- Built and maintained Magento online stores for multiple clients with PHP and JavaScript, from concept to deployment; stores live in production include World Combat, Wilson Atacado, Mirão and Cama in Box.

- Integrated SOAP APIs and XML payloads to connect stores with external ERPs and third-party systems.
- Implemented technical SEO and performance optimization on stores with real traffic; delivered custom integrations end to end.

SELECTED PROJECTS

- **SlotForge — .NET 8 slot engine and Monte-Carlo RTP simulator**
github.com/PedroDelgadoHenriques1/SlotForge
Server-side (RGS-style) slot engine in C#: cryptographic and seeded RNG without modulo bias, integer-only money math, payline/wild/scatter evaluator and free spins. Simulator runs 5M rounds at ~228k spins/s and tuned the sample game from 157% to 97.65% RTP. ASP.NET Core Minimal API, 30 xUnit unit and integration tests, Docker, GitHub Actions CI.
- **FIAP Cloud Games — OrchestrationApi (postgraduate capstone)**
github.com/FIAP-PosTech-2025/OrchestrationApi
Games platform across 5 .NET 8 microservices: Kong API Gateway (JWT, rate limiting, CORS), RabbitMQ, Redis, SQL Server, MongoDB, AWS Lambda (LocalStack + Terraform), Prometheus + Grafana; runs end to end with Docker Compose or Kubernetes.
- **Unity Technical Art — URP Stylized Shaders and Texture Optimizer**
github.com/PedroDelgadoHenriques1/urp-stylized-shaders
Four real-time URP shaders handwritten in HLSL with procedural noise from scratch; Editor tool that scans textures, estimates runtime memory and batch-applies compression fixes, reporting the exact memory saved.

EDUCATION

Postgraduate Degree in .NET Systems Architecture (Pós Tech, 360 h) FIAP + Alura	<i>Oct 2025 – Oct 2026 · in progress</i>
Bachelor's Degree in Software Engineering UniSenaiPR — Faculdade da Indústria	<i>Jul 2021 – Jun 2025 · completed</i>

LANGUAGES

Portuguese (native) · English (advanced, B2 — technical reading, writing and meetings)